Forward Propagation

Forward Propagation: How Neural Networks Predict

1 The Idea

Forward propagation means **sending input data forward through the network** layer by layer until we get the final prediction.

Think of it like a factory pipeline:

- Raw material (input features) enters.
- Each machine (layer of neurons) processes it.
- At the end, we get the finished product (prediction).

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2 Steps of Forward Propagation

(a) Input Layer

- You start with input features. Example:

X = [x1 , x2 , … , xn]

- These are passed into the network.

(b) Weighted Sum

- Each neuron computes a **linear combination** of inputs:

$$z_j^{[l]} = \sum_i w_{ij}^{[l]} a_i^{[l-1]} + b_j^{[l]}$$

- Here:
  - w = weights (how strong each input is).
  - b = bias (a shift term).
  - $a^{[l-1]}$ = outputs of previous layer (or input x if it's the first layer).

(c) Activation Function

- After computing z, we apply an **activation function** (sigmoid, ReLU, tanh, softmax, etc.).

$$a_j^{[l]} = g(z_j^{[l]})$$

- This adds **non-linearity**, helping the network learn complex patterns.

(d) Pass Forward

- The activations $a^{[l]}$ of one layer become the input for the next.

(e) Output Layer

- Finally, the output layer produces predictions:
    1. For classification: probabilities (e.g., cat vs dog).
    2. For regression: a number (e.g., house price).

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3 Example

Say we have:

- Inputs: $x_1 = 2, x_2 = 3$
- One hidden layer with 2 neurons
- One output neuron

Step 1: Hidden Layer

$$z^{[1]} = W^{[1]}x + b^{[1]}, \quad a^{[1]} = g(z^{[1]})$$

Step 2: Output Layer

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}, \quad a^{[2]} = g(z^{[2]})$$

That $a^{[2]}$ is the prediction.

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4 Intuition

- Inputs = raw ingredients.
- Weights = recipe instructions (how much of each ingredient to use).
- Bias = secret spice (extra adjustment).
- Activation = cooking method (raw → delicious dish).
- Output = final meal (prediction).

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5 Why It Matters

- Forward propagation is **used in prediction (inference)**.
- During **training**, we combine forward propagation + loss calculation + backward propagation (to update weights).

In short:

Forward propagation =

Inputs → Weighted sum + bias → Activation → Next layer → … → Final prediction.
